



MANPREET KAUR

Registration Number: 20233180

Computer Science and Engineering

Motilal Nehru National Institute of Technology Allahabad

+91-7819081686

✉ kaurmanpreet2k27@gmail.com

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EDUCATION

- Motilal Nehru National Institute of Technology Allahabad** 2023 - Present
Bachelor of Technology in Computer Science and Engineering CPI: 8.14 (till 5th sem)
- NBPS Hastinapur, Meerut** 2022
Class XII, Central Board of Secondary Education Percentage: 98.2%
- NBPS Hastinapur, Meerut** 2020
Class X, Central Board of Secondary Education Percentage: 93.2%

PROJECTS

- PortBiter** Mar 2026 – April 2026
Autonomous AI multi-agent web security scanner leveraging LLM-driven orchestration to dynamically detect vulnerabilities and generate actionable security reports.
 - **Tech Stack:** Python, FastAPI, LangGraph, Groq (LLM), Multi-Agent Systems, SQLite, Next.js, WebSockets, Docker
 - Built a **multi-agent AI system** (planner, executor, analyzer) reducing manual security testing effort by **70%** through adaptive, context-aware decision making.
 - Designed a **strict tool registry and policy validation layer** to prevent hallucinated actions, ensuring **safe and reliable agent-tool execution**.
 - Implemented **stateful orchestration** using LangGraph with checkpointing, enabling fault-tolerant, iterative, and resumable scan workflows.
 - Developed **real-time scanning infrastructure** with FastAPI and WebSockets, providing live progress tracking, logs, and vulnerability insights.
 - Engineered **multi-vector vulnerability detection** (XSS, SQLi simulation, misconfigurations, file exposure) with structured reporting, improving remediation efficiency by **40%**.
- Kharido.in** February 2025 – March 2025
Full-stack e-commerce platform with integrated secure payment systems, dynamic product catalog management, and enhanced user engagement features.
 - **Tech Stack:** MongoDB, Express.js, React, Node.js, Stripe API, Razorpay API, JWT, Redux, Material-UI
 - Implemented **Google OAuth authentication**, reducing login time by 60% while improving security.
 - Integrated both **Stripe and Razorpay payment gateways** to provide customers with flexible and secure transaction options.
 - Designed and deployed a real-time **product review system** that increased user engagement by 75%
 - Built an efficient **admin dashboard interface** with role-based access controls.
 - Added comprehensive **order tracking** with status updates, cutting support queries by 50%.
 - Developed an **advanced search system** featuring category-wise filters, price sorting (high-to-low/low-to-high), and keyword search that improved product discovery rates by 40%.
 - Optimized frontend performance using **Redux state management** and efficient component rendering.

TECHNICAL SKILLS

Programming Languages: C++ , C

AI / LLM & Agentic Frameworks: FastAPI, LangChain, LangGraph, MCP Server, Agentic AI Systems

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Redux Toolkit

Databases: MongoDB, SQLite

Developer Tools & Platforms: VS Code, GitHub, Postman, MongoDB Compass, Docker

Core CS Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Operating Systems, DBMS, Computer Networks

ACHIEVEMENTS

- Ranked 1** in **Codesparks**, a competitive coding contest with 150+ participating teams 2023
- Ranked 2** in **HackQuest**, Machine Learning Hackathon showcasing applied ML solutions 2023
- Ranked 6** in **Codesangam**, a Web Development competition recognized for full-stack application design 2024
- Ranked 6** in **Hack36**, a 36-hour hackathon themed around **Cybersecurity** 2025
- Ranked 3** in **Hack 18**, an 18-hour hackathon, for building an **agentic AI** travel planning system 2026

CODING PROFILES

LeetCode: [_m_kaur1905_](#)

Codeforces: [Kaur_manpreet](#)

POSITIONS OF RESPONSIBILITY

- Mentor** Student Mentorship Program, MNNIT Allahabad August 2024 – Present
- Web Developer** Enactus, MNNIT Allahabad September 2024 – Present
- Web Developer** Girl Up Coders, United Nations Foundation (UNF) initiative June 2025 – Present